Same Story, Different Fate: Context, Content and Chance in Online News Popularity

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Abstract

Popularity models for the Mashable news benchmark reach a ROC-AUC of about 0.74, but it is unknown what that skill consists of: knowing *when* and *where* an article appears, *what story* it covers, or *how* it is written. The benchmark ships no text, yet every article's URL ends in a shortened headline. Retrieving near-identical headlines published within a week, we find 2,941 pairs of same-story "twin" articles (78% precision on 180 annotated pairs) and use them as a topic-controlled natural experiment. With strictly past-only models we find that (i) publication time contributes no ranking skill, the editorial channel about 0.045 AUC, and fixing the story no detectable further amount: within a channel and day the model still ranks at 0.675; (ii) same-story articles barely share their fate (residual correlation 0.05), and the earlier twin gets about 7% more shares, a sign that follow-ups split attention; (iii) of 13 content features, only the subjectivity of the body survives topic control, and it more than doubles (0.07 to 0.17 log shares per sd); (iv) articles fare worse on crowded days (doubling daily volume: −18% shares; also within twins, not in a placebo). Since much remains chance, we show what a model can honestly promise: 80% intervals spanning a factor of six, kept calibrated under drift by adaptive conformal inference, and 87% accuracy on the 10% of articles it is most sure about.

1 Introduction

Editors decide every day which of dozens of new articles to promote, and popularity prediction before publication is meant to support that decision. The Online News Popularity dataset [6] is the standard benchmark: 39,644 Mashable articles with 58 pre-computed features and their share counts. Work on it mostly compares classifiers on the given features; tree ensembles reach an AUC of about 0.73–0.74.

A single AUC hides *what* is being predicted. An article's shares depend on its context (the day, the section it is placed in, what else appears that day), on the news story it covers, on how it is written, and on chance; experiments on cultural markets show that the last part can be large [15, 19]. If a model mainly recognises popular sections, it says little to an editor choosing between two pieces on the same story. Separating these sources needs comparisons in which the story is held fixed, which observational data rarely offer.

We find such comparisons inside the benchmark. Mashable URLs end in a slug, a shortened headline (*/2013/01/13/aaron-swartz/*). Articles whose slugs nearly coincide and that appear within a week usually cover the same story (Table 2). These *twins* turn the dataset into a topic-controlled natural experiment in the spirit of paired designs on Twitter [22] and Reddit [14], which, to our knowledge, have not been applied to this benchmark.

Research questions.

RQ1 Where does predictive skill live: in time, channel, story or the article itself? Do same-story articles share their fate?

RQ2 Which content effects survive when the story is held fixed?

RQ3 Does the publishing context matter beyond the article: competition from other articles that day, and the recent popularity of similar stories?

RQ4 Given what remains unpredictable, what can a model honestly promise under temporal drift?

Our contributions are a twin-retrieval method with a measured precision; a past-only decomposition of AUC into nested contexts; topic-controlled estimates of content and competition effects with formal tests against the naive estimates; and calibrated uncertainty under drift.

2 Related Work

Popularity prediction. Fernandes et al. [6] introduced the dataset and reached an AUC of 0.73 with a random forest on rolling windows. Prediction before publication is hard [1]; early share counts help much more [21], but are not available here.

Limits to prediction. In artificial music markets the same song succeeded or failed depending on social influence [19], and even with full information cascade sizes on Twitter are only partly predictable [15]. Hofman et al. [12] argue for reporting what is predictable and from what. We measure this on news, decomposing skill by context.

Table 1: Feature groups (60 columns incl. two cleaning flags).

Group	#	Examples
Content	12	words, links, images, videos
Keyword (`kw_*`)	10	shares of earlier keyword articles
Self-reference	3	shares of linked articles
Channel	6	lifestyle, entertainment, business, ...
Weekday	8	day of publication, weekend
LDA topics	5	closeness to 5 topics
Sentiment	16	polarity, subjectivity, word rates

Controlled comparisons. Tan et al. [22] compared tweets by the same author on the same link; Lakkaraju et al. [14] compared resubmissions of the same image on Reddit; headline A/B tests give randomised versions [16]. Our twins are a weaker, observational analogue: same story, but different articles.

Attention and uncertainty. Items compete for limited attention [24, 25]. Conformalised quantile regression [18] gives distribution-free intervals; adaptive conformal inference [11] keeps coverage under drift [9]; selective classification [10] predicts only when confident. Work on learning from streams of networked items [26, 27] motivated our strictly past-only evaluation.

3 Data and Twin Articles

Data. 39,643 articles (one row with an impossible token rate removed) published between 7 January 2013 and 27 December 2014, with 58 features (Table 1) and shares recorded once, in January 2015. We parse the date and slug from the URL. Shares are heavy-tailed (median 1,400, mean 3,395, maximum 843,300); we model log(1+shares) and call an article *popular* if it has at least 1,400 shares [6]. Two feature groups are built from other articles' shares: keyword statistics (`kw_*`) and statistics of linked Mashable articles. The keyword statistics use only articles from earlier days (`kw_max_max` equals the running maximum of earlier shares for 99.87% of articles), but take those articles' shares from the collection date, so their gains are likely optimistic, and three of them mostly encode the date.

Twins. We represent slugs by TF-IDF vectors [20] of word uni- and bigrams and call two articles twins if their cosine similarity is at least 0.5 and they appeared at most 7 days apart (3,382 pairs). Some of these are instalments of recurring columns (*viral video recap*, *olympics day four* vs. *day eight*, episode recaps), which a rule-based filter removes (identical slugs that recur at least four times or are ≥5 days apart, slugs differing only by numbers or ordinals, both containing "recap"), leaving 2,941 pairs in 1,278 stories (connected components of the twin graph).

To measure precision we annotated 180 pairs, 30 per similarity band from 0.3 to 1.0, as same story, series instalment, same broad topic, or unrelated, from slug, dates and channel. The annotation was done by the authors' assistant, not by independent annotators. The

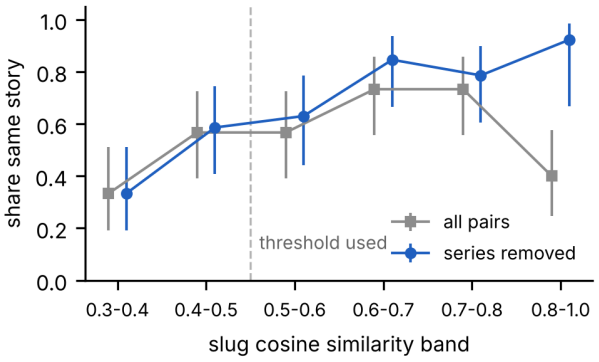


Figure 1: Share of annotated pairs that cover the same story, by slug similarity (30 pairs per band, 95% Wilson intervals), before and after removing recurring-series pairs.

Table 2: Four same-story twins (slug words, shares).

Earlier article		Later article	
aaron swartz	7,900	aaron swartz petition	581
game of thrones	2,600	game of thrones trailer	22,300
samsung galaxy s4 mini	6,300	galaxy s4 mini	920
typhoon hagupit philippines	4,700	typhoon hagupit	1,100

filter caught 27 of 33 series pairs with no false positives. After filtering, pairs with similarity ≥0.5 are the same story in 78% [68, 85] (Wilson) and the same story or topic in 96%. Without the filter, the most similar band is mostly series (Fig. 1), so a higher threshold alone does not give cleaner twins. Twins are rarely in the same channel (29%): channel labels agree only loosely with slug topics (e.g. 27% of *game of thrones* slugs are labelled entertainment), so we treat channel as an editorial placement rather than a topic.

4 Methods

Benchmark models. Logistic regression (L1), k-NN, random forest and histogram gradient boosting (HistGB) [3, 4, 8, 13, 23], tuned on a chronological validation block (oldest 70% train, next 10% validation, newest 20% test) in scikit-learn [17].

Past-only scores. All RQ1 analyses use scores from models trained only on earlier articles: for every month from July 2013 on, we fit HistGB (and L1 logistic regression) on all articles of earlier months and score that month (30,446 scored articles).

Pairwise AUC within a context. For a grouping G, the within-G AUC is the probability that a popular article scores higher than an unpopular one *from the same group*. With one group it is the usual AUC; fixing finer groups (week, day, channel, channel×day) removes the skill that only tells groups apart. For twins we use the pairs themselves. Because the 1,400 cut makes many twin pairs "discordant" by a few shares, we also report concordance on pairs whose shares differ by more than a factor of two. Intervals are cluster bootstraps [5] over publication weeks, or over stories for twins.

Twin designs. Do twins share their fate? We correlate residual log shares (after channel×week and weekday fixed effects) of the two twins and compare with a matched neighbour: the second twin replaced by a random article from its channel and day. For content effects (RQ2) we regress the within-pair difference (earlier minus later) in log shares on differences in 13 standardised content features, a constant (the earlier-article advantage) and channel-difference dummies, and compare with a cross-sectional regression with channel×week and weekday fixed effects. The difference between the two estimates is tested by combining independent bootstraps; twin coefficients are corrected for multiple testing [2].

Competition, memory and uncertainty. RQ3 regresses log shares on the log number of articles in the dataset that day, with week, weekday and channel fixed effects and day-clustered errors, and checks a placebo (next week's volume on the same weekday), content controls and twins. A *semantic memory* feature averages the log shares of the 20 most similar earlier slugs with time-decay weights $0.5^{\text{age}/h}$.[1] RQ4 fits 10% and 90% quantile HistGB models, conformalises them on the validation block (CQR) [18], and adds adaptive conformal inference (ACI, step 0.005) [11]; selective prediction keeps the articles with the largest $|p - 0.5|$.

Reproducibility. Seeds are fixed; every number and figure comes from a script in the code, and `run_all.sh` reproduces all of them from the raw CSV.

5 Results

Benchmark. On the chronological test block (Table 3) random forest and HistGB tie (gap 0.001 [−0.003, 0.005], weekly block bootstrap) and beat L1 logistic regression by 0.020 [0.014, 0.025]. The share-derived features are worth 0.034 [0.027, 0.040] AUC. Ranking holds up under a chronological split, and a later dip (Oct–Dec 2014) reflects harder articles more than an outdated model: retraining up to October recovers only 0.006 [0.003, 0.010]. Training-set size and regularisation matter only for small samples: random forest is best below 10,000 training articles, and with 1,000 articles strong L1 regularisation gains 0.03 AUC, while with all data the penalty hardly matters.

5.1 Where does the skill live? (RQ1)

Time contributes nothing, channel some, story nothing detectable. Past-only HistGB ranks all scored articles at 0.734 [0.725, 0.740] (Fig. 2). Comparing only articles from the same week (0.734) or day (0.727) barely changes this, so the model does not profit from knowing which days were good. Comparing within a channel costs 0.045 (0.690 [0.682, 0.696]), and within channel and day 0.675 [0.668, 0.683]. Holding the story fixed adds no detectable loss: same-story twins are ranked at 0.719 [0.692, 0.748], and same-story twins in the same channel at 0.672 [0.602, 0.738], overlapping the channel-and-day level. The pattern is the same for logistic regression (0.709 to 0.639) and for HistGB without share-derived features (0.701 to 0.628), and for concordance on pairs that differ more than twofold (0.714 overall, 0.667 within channel and day, 0.690 [0.656, 0.727] within twins). Most of what the model knows is therefore about the article, not its context: it separates two articles on the same story about as well as two random articles from the same section and day.

Table 3: Chronological test results (7,929 newest articles), threshold 1,400 shares; settings chosen on the validation block; 95% weekly block-bootstrap intervals.

Model	ROC-AUC	PR-AUC
All features (60)		
L1 LogReg	0.717 [.701,.733]	0.696
kNN (k=500)	0.705 [.688,.722]	0.682
Random forest	0.736 [.721,.751]	0.715
HistGB	**0.737** [.721,.752]	**0.719**
Without share-derived features (47)		
L1 LogReg	0.693 [.677,.710]	0.670
Random forest	**0.708** [.694,.722]	**0.691**
HistGB	0.703 [.687,.720]	0.683

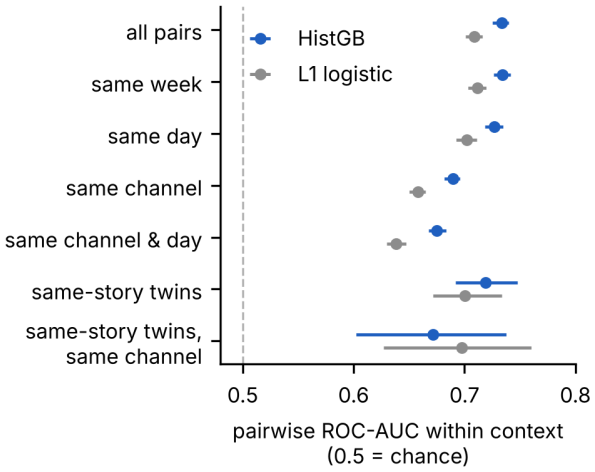


Figure 2: RQ1: pairwise ROC-AUC of past-only scores among articles that share a context (95% cluster-bootstrap intervals: weeks, or stories for twins).

Same story, different fate. Twins hardly share their popularity. Their residual log shares correlate at 0.051 [0.016, 0.086], against 0.002 for matched random neighbours; within a channel at 0.103 [0.040, 0.167] (Fig. 3a). The story thus accounts for roughly 5% of the variance that week and channel leave, which is itself small: week, channel×week and channel×day means explain 1.5%, 7.8% and 9.7% of the variance of log shares. The earlier twin gets 0.070 [0.005, 0.132] more log shares (about 7%), whereas the earlier of two random same-week articles does not (−0.002); for the most similar twins (cosine ≥0.7) the advantage is 0.135. A follow-up

[1] The time decay was inspired by the TimeDecay mechanism of the open-source VCPToolBox project (`https://github.com/lioensky/VCPToolBox`); our implementation is independent.

on the same story competes with its predecessor, so the low correlation may partly reflect split attention rather than an irrelevant story; the two readings cannot be separated with one snapshot of shares (Section 6).

5.2 Content with the story held fixed (RQ2)

Most content effects that look clear across articles cannot be confirmed within twins, because twin differences are noisier: the two articles of a pair differ widely in format (their body lengths correlate at 0.02). One effect survives and grows (Fig. 3b): a one-sd more subjective body goes with 0.070 [0.057, 0.084] more log shares across articles but 0.166 [0.092, 0.239] within twins (BH-adjusted $q = 0.013$); the difference, 0.097 [0.018, 0.165], excludes zero. It holds for twins with cosine $\geq$0.6 (0.149), within the same channel (0.122) and for twins of similar length (0.165). An opinionated take on a story thus beats a neutral one on the same story more than the cross-sectional estimate suggests. Links (0.067 [0.010, 0.132]) and average word length (-0.113 [-0.198, -0.030]) agree in sign with the cross-section but do not pass the multiple-testing correction. The video effect (0.054 [0.041, 0.068] across articles) shrinks to 0.007 [-0.041, 0.059] within twins (difference $p = 0.09$), suggesting that video-heavy stories rather than videos are popular. Negative tone, measured as the share of negative words, has no detectable effect in either design once subjectivity and positive words are controlled. Across 30 near-equally accurate models (validation AUC within 0.01) the ranking of feature groups by permutation importance is stable at the extremes (keywords first, sentiment last) [7].

5.3 Publishing context: competition and memory (RQ3)

Articles fare worse on crowded days (Fig. 4a). With week, weekday and channel fixed effects, a log unit more articles that day goes with -0.286 [-0.372, -0.200] log shares: doubling the volume (days range from 12 to 105 articles, mean 55) means 18% fewer shares. The estimate is unchanged by all content controls (-0.282), holds for long articles (-0.216), without the busiest 5% of days (-0.309), within twins (-0.200 [-0.324, -0.076]) and for both same-channel (-0.108) and other-channel volume (-0.153), while the placebo is small and not detectable (-0.059 [-0.131, 0.014]). It is three to four times larger in 2013 than in 2014. We read this as competition for limited attention [24], but it remains an association: busy days may differ in ways we do not observe.

The popularity of similar earlier stories does not help. Semantic-memory features correlate with log shares at only 0.016–0.029 for every half-life and change past-only AUC by -0.0008 [-0.0019, 0.0004]; recency-weighted training did not help either. What matters is the article and its day, not what similar stories did before.

5.4 What can a model promise? (RQ4)

An 80% conformal interval for a typical test article runs from 776 to 5,017 shares, a factor of 6.2. Calibrated once on the validation block, it covers 79.5% of the test articles overall but only 76.3% in September 2014; ACI keeps every month between 79.6% and 81.0% (Fig. 4b), assuming shares become known before the next article is scored, which is optimistic. Viral articles are hardest: at the 95th-percentile threshold ROC-AUC is unchanged (0.73) but PR-AUC falls to about 0.10, although a weekly shortlist of the top 5% still contains three times the base rate of viral articles (12% [9, 15] vs. 4%). A model can still be useful when it is allowed to abstain: past-only accuracy is 0.673 on all articles, 0.785 on the 40% it is most sure about and 0.869 on the top 10% (Fig. 4c).

6 Discussion

What the 0.74 is made of. The benchmark's AUC is mostly article-level skill. Time contributes nothing, the editorial channel about 0.045, and the story nothing we can detect. This changes how to read the benchmark: a better model is not one that recognises popular sections but one that separates articles that compete for the same readers. Within such pairs the most robust content signal is subjectivity of the body.

Story irrelevance or split attention? Same-story articles correlate weakly. Two readings fit: the story matters little, or it matters but its audience is split between articles, which would also lower the correlation. The earlier-twin advantage and the volume effect support some splitting. Separating them would need share histories or randomised versions [16]; with one snapshot we report the twin correlation as an upper bound on how much "the story" alone tells, given this splitting.

Limitations. Twins are found from slugs only and are 78% precise; the remaining same-topic pairs bias correlations and effects towards cross-sectional values. Annotation was not independent. Twin differences remove the story and the week but not everything else (an editor may give a scoop more prominence). The daily volume counts articles in this dataset, which may not equal Mashable's true output. Shares were recorded once, so age and publication date cannot be separated, and keyword statistics use post-publication shares of earlier articles. All results come from one outlet in 2013–2014.

Advice for work on this dataset. Evaluate chronologically with past-only models; report within-context AUC next to pooled AUC; treat cross-sectional content effects as topic-confounded until checked with twins; and report calibrated intervals or selective accuracy, since much of popularity is not predictable from these features.

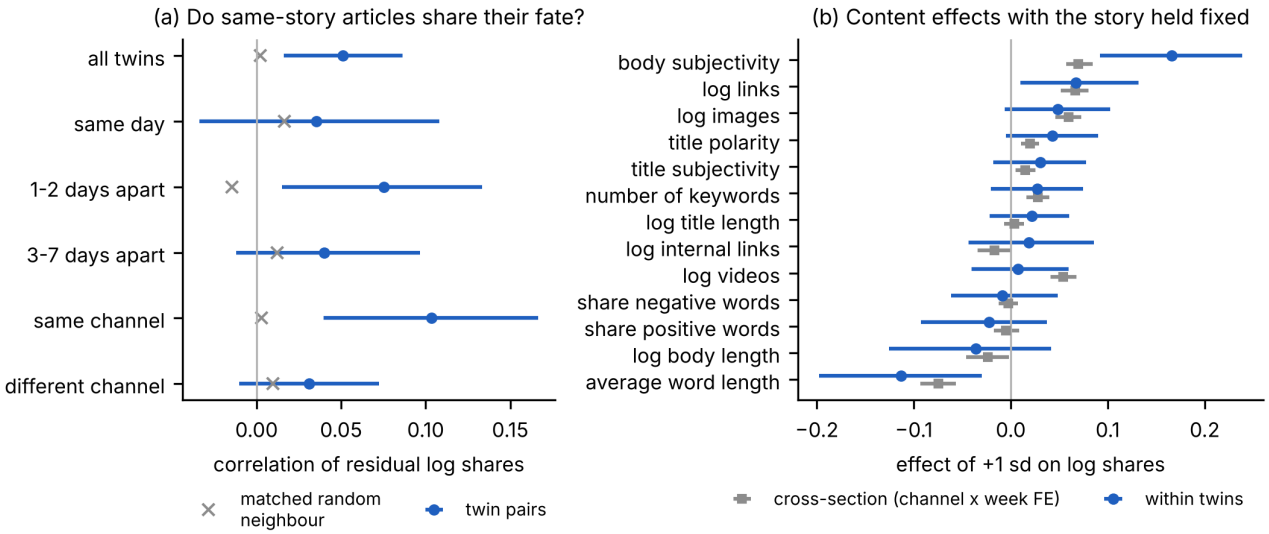


Figure 3: (a) RQ1: correlation of residual log shares (after channel×week and weekday effects) between twins and between matched random neighbours from the same channel and day, by subset (95% story-cluster intervals). (b) RQ2: effect of a one-sd increase in each content feature on log shares, across articles (channel×week fixed effects, week-cluster intervals) and within same-story twins (story-cluster intervals).

7 Conclusion

Hidden in the URLs of a standard benchmark are thousands of same-story twins. They show that news popularity is predictable mostly at the level of the individual article (RQ1), that same-story articles share little of their fate and that the first one has an edge (RQ1), that only body subjectivity among content features survives topic control and even grows (RQ2), that articles lose shares on crowded days while similar past stories carry no memory (RQ3), and that honest predictions are wide but can be kept calibrated under drift and sharpened by abstaining (RQ4).

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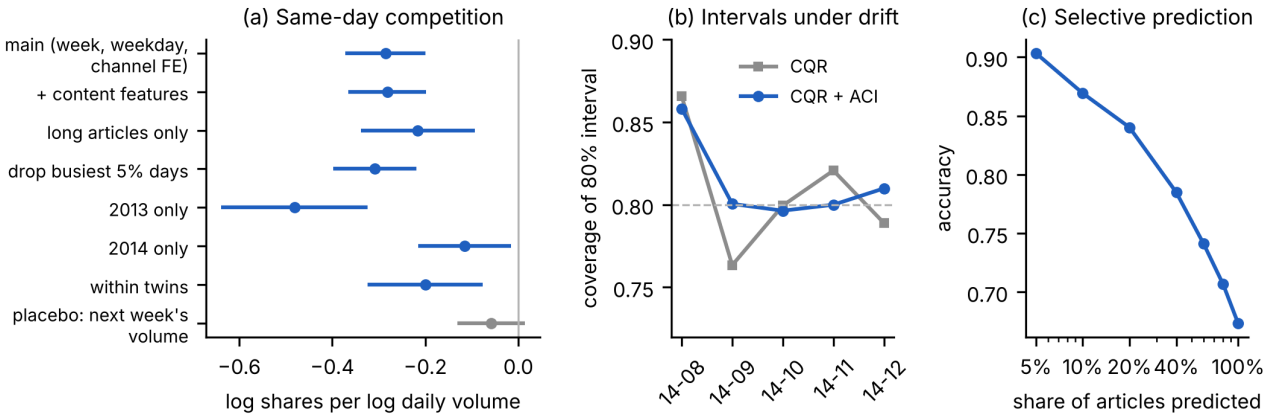


Figure 4: (a) RQ3: change in log shares per unit of log daily volume (95% day- or story-clustered intervals); the placebo uses the volume of the same weekday one week later. (b) RQ4: monthly coverage of 80% intervals on the test block, conformalised once (CQR) or adaptively (ACI). (c) RQ4: accuracy when predicting only the given share of articles with the most confident past-only scores.

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